VEDANSH AIREN

Full-Stack Developer • Applied AI for Enterprise Workflows • Data & Process Automation

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SUMMARY

Full-stack developer with an M.S. in Computer Science and 5+ years of software development across the whole path from schema to screen: relational databases, REST APIs, server-side PHP and Node.js, React front ends, and Python automation. For the past two years at University of Iowa ITS, I have applied AI to real institutional work. I sit with the staff whose process is about to change, then build the automation, grounded agents, and accuracy tests that make it dependable, for 8+ units and 1,000+ users. I am the service owner for Claude, ChatGPT Edu, and Microsoft 365 Copilot and led a 147-person pilot of a secure multi-model AI platform. Comfortable with messy data, patient health and institutional data rules, research teams, and non-technical audiences.

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, PHP, SQL, C#, Java

Web & APIs: React, Node.js/Express, PHP MVC (CodeIgniter), .NET, Angular; REST API design and integration; WordPress (custom themes and plugins), Drupal; HTML/CSS, Bootstrap, Tailwind

Data & Reporting: MySQL, Microsoft SQL Server, MongoDB, DynamoDB; schema design, migrations, query optimization, data validation; reporting dashboards, Excel/Power Query

Applied AI & Automation: OpenAI, Anthropic Claude, Amazon Bedrock, and Whisper APIs; LibreChat; retrieval-augmented generation (RAG) and prompt engineering; Microsoft 365 Copilot and Copilot Studio agents; ChatGPT Edu Custom GPTs; Gravyty/Ivy.ai; Power Automate; AI accuracy evaluation and regression testing

Identity, Cloud & DevOps: Auth0 (Okta) and AWS Cognito authentication, SSO; AWS (EC2, Cognito, DynamoDB), Azure, GCP, DigitalOcean; Docker, Kubernetes; Git (GitHub Actions, GitLab CI, Bitbucket); Unix/Linux

Practices: Code review; unit, integration, and end-to-end testing (Jest, Jasmine, pytest); Agile/Kanban in Jira; requirements gathering; technical documentation and SOPs; AI-assisted development (Claude Code, OpenAI Codex)

EXPERIENCE

Emerging Technology Strategist (AI Developer & Service Owner) **Nov 2024 – Present**

University of Iowa Information Technology Services, Office of the CIO — Iowa City, IA • formerly titled AI Support Team Consultant

- **Process automation:** mapped a manual certificate workflow with HawkAI program staff, then built a Python and Power Automate pipeline that cleans attendance and grade data and generates and delivers certificates; 1,120 issued, no manual steps.
- **Grounded AI agents and chatbots:** interviewed department staff to document how their work really runs, then designed, built, and maintain chatbots and Copilot agents grounded in each unit's approved knowledge sources for 8+ units and 1,000+ users (OpenAI API, Copilot Studio, Gravyty/Ivy.ai), including HR and OneIT pilots that deflect routine questions.
- **Catching wrong answers:** built testing infrastructure that measures AI answer accuracy and flags regressions; ran black-box evaluations of enterprise AI tools before campus adoption; wrote ITS guidance that positions agents as support tools requiring human review, not final decision-makers.
- **Iowa AI Gateway:** led the pilot of a secure, HawkID SSO multi-model AI workspace (LibreChat; OpenAI, Anthropic, and Amazon Bedrock models) for 147 faculty, staff, and researchers; turned use cases into configuration (agents, artifacts, custom model parameters) and used pre/post-pilot survey analysis to inform the production decision.
- **Service ownership:** service owner for Claude, ChatGPT Edu, Microsoft 365 Copilot and Copilot agents, and Gravyty chatbots; advise faculty and staff on licensing, data handling, and compliance questions; drafted the AI tool onboarding lifecycle (Evaluate, Request & Approve, Fund, Provision, Support, Sustain) and am building the enterprise Claude rollout roadmap; troubleshoot platform issues with vendors; helped move Microsoft 365 Copilot from pilot to production.
- **Communication:** trained 400+ faculty, staff, and students ("Code with AI," Copilot Agents, AI Gateway); helped deliver HawkAI courses to 1,500+ learners; published three ITS articles on AI agents and Drupal knowledge pages for Microsoft 365 Copilot.

Software Engineer (Part-Time) **Feb 2026 – Jul 2026**

FarrPro, healthcare software — Iowa City, IA

- **Titronics Cloud:** built features for a patient portal serving thousands of patients (React, Node.js/Express APIs, MySQL, AWS EC2 and Cognito): secure scan access, provider messaging, provider networks, and Stripe payments.
- **MyCare:** developed an AI health-records app (React, TypeScript, Vite, Node.js, Auth0) with Whisper transcription and HL7 FHIR integration; supported a 100+ user beta.
- **Titronics:** maintained a .NET/C# and MySQL clinical thermography measurement system used by 100+ clinics.
- Reviewed teammates' code and ran integration and end-to-end tests on every new service before release (Whisper API, Google Maps Platform API) across three concurrent products, using Jira/Kanban, Claude Code, and Codex.

Graduate Research Assistant, then AI Developer **Aug 2022 – Nov 2024**

Holo Reality Lab, Electrical & Computer Engineering, University of Iowa • RA Aug 2022 – May 2024; AI Developer Aug – Nov 2024

- As AI Developer, led development of a generative AI chatbot to support a gerontology research group and improve its access to scientific knowledge.
- Built an AI-driven VR classroom (Unity, C#, OpenAI API) with real-time heart-rate-variability monitoring for a human-subjects teacher-training study, raising study engagement 38%; co-authored the resulting IS&T Electronic Imaging 2025 paper.
- Analyzed tracking and physiological study data in MATLAB and Python, documenting methods for reproducibility.

Software Engineer **Jul 2021 – Jul 2022**

Chenoa Information Services, healthcare IT — India

- Built patient registration, appointment scheduling, and provider dashboard modules for a hospital web app (Angular, Node.js, DynamoDB), improving scheduling efficiency 45% and usability 50%.
- Integrated the Google Vision API so insurance details were captured from a scan of the patient's card instead of typed by hand, cutting intake processing time 70%.
- Led a legacy-to-modern Microsoft SQL Server migration, writing optimized queries and validating data integrity; data access 35% faster. Delivered in Agile sprints with Jira and Bitbucket (Git) at a 98% sprint resolution rate.

Software Developer **Sep 2020 – Jun 2021**

Simplex Metal Processors, manufacturing — Indore, India

- Moved manual operations into a PHP (CodeIgniter MVC) and MySQL web application for cost calculation, financial tracking, quality checks, and automated spreadsheet reports; designed the database and deployed on DigitalOcean to 7 companies, cutting manual tasks 50% and errors 30%.
- Documented workflows, queries, and user guides; provided first-tier support and training through adoption.

Web Developer Intern **Jun 2019 – Aug 2019**

Chetaru Web Link — Indore, India

- Built custom WordPress templates and plugins for 10+ client sites (WooCommerce, Elementor, Jetpack) with weekly updates.

SELECTED PROJECTS

- **EvalForge** (Python, FastAPI, asyncio, Next.js, Docker, Kubernetes, GitHub Actions): multi-LLM evaluation and regression-testing service that scores Anthropic, Google Gemini, and OpenAI models on correctness, groundedness, latency, and cost with a validated LLM judge; 72 live evaluations with zero request or judge errors, and a 4-job CI pipeline running 87 tests.
- **Hospital Management System** (React, Node.js/Express, MySQL, Jest): schema to screen, from the relational schema (ER model) through a layered REST API to role-based dashboards for patients, providers, and admins.
- **Voting Web Application** (Node.js/Express, MySQL, Jasmine): admin workflows for election setup, booth configuration, and voter approval; GitLab CI pipeline with SonarQube code-quality configuration.
- **Simplex Metal Processors website** (WordPress, 2019–20): custom child theme with interactive 3D models and Google Maps that lifted site traffic 15x.

PUBLICATIONS & TECHNICAL WRITING

- L. Flores, S. King, V. Airen, T. Bell, "Enhancing teacher training with AI-guided simulations in smart virtual reality," Engineering Reality of Virtual Reality, IS&T Electronic Imaging 2025 (ERVR-163).
- University of Iowa ITS articles (2026): "Copilot agents as cognitive partners," "AI agent vs. AI chat," and "Build once, reuse often: Comparing custom AI assistants."

EDUCATION

M.S. in Computer Science, University of Iowa • GPA 3.84/4.0 **May 2024**

Coursework: Database Systems, Software Engineering & Architecture, Cloud Computing, Generative AI, Software Architectures for Healthcare Information Systems, Research Ethics

B.Tech in Computer Science & Information Technology, Symbiosis University of Applied Sciences, India **May 2021**

CERTIFICATIONS & LEADERSHIP

- **CITI Program:** Responsible Conduct of Research; IRB Group 2 (Social & Behavioral Research); Export Controls.
- **LinkedIn Learning:** Getting Started with Microsoft Azure; Azure Essential Training for Developers; Power BI Essential Training.
- President, Computer Science Graduate Student Association (50+ members), University of Iowa, 2023–24.
- Winner, IdeaStorm Pitch Competition, University of Iowa, 2023.